MILITARY SPECIFICATION

TEST HARNESS, RADAR, AN/APM-293A

This specification has been approved by
Naval Air Systems Command, Department of the Navy.

1. SCOPE

1.1 Scope - The Test Harness, Radar AN/APM-293A, hereinafter referred to as the equipment, covered by this specification shall be for intermediate level use and shall facilitate testing and maintenance of Radar Set AN/APQ-124, or AN/APQ-124A. The equipment shall include the items necessary to position, support, and interconnect the componenets of Radar Set AN/APQ-124, or AN/APQ-124A and associated test equipment when operating on the bench.

1.2 Classification - The equipment covered by this specification shall consist of the following items:

Item	Type Designation	Para.
Test Stand	MT-2579/APM-145	3.3.1
Base, Electrical Equipment	MT-4046A/APM-293	3.3.2
Base, Electrical Equipment	MT-4048/APM-293	3.3.3
Base, Electrical Equipment	MT-4047/APM-293	3.3.4
Base, Electrical Equipment	MT-4049/APM-293	3.3.5
Panel, Electrical Equipment	MT-4045/APM-293	3.3.6
Base, Hydraulic Pump	MT-4050/APM-293	3.3.7
Mounting	MT-2307/APM-145	3.3.8
Mounting	MT-2308/APM-145	3.3.9
Mounting	MT-3249/APM-145B	3.3.10
Maintenance Fixture, Electronic	MT-3973/ASA-63	3.3.11
Interconnecting Box	J-2826A/APM-293	3.3.12

FSC 4940

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Hose Assembly, Hydraulic	MX-3012/APM-145	3.3.13
Hose Assembly, Hydraulic	MX-3571/APM-145A	3.3.14
Hose Assembly, Hydraulic	MX-3572/APM-145A	3.3.15
Lift, Transfer, Radar Set	MX-8196/APM-293	3.3.16
Adapter, Plate, Radar Antenna	MX-4844/APM-197	3.3.17
Mount, Gyroscope	MT-2093/APQ-83	3.3.18

1.3 Associated Equipment - This equipment shall operate with the associated equipment listed in 6.9.

2. APPLICABLE DOCUMENTS

2.1 General - The following documents of the issue in effect on the date of invitation for bids form a part of this specification to the extent specified herein.

SPECIFICATIONS
Military

MIL-T-(To be assigned)	Radar Set AN/APQ-124A, Military Specification for
MIL-I-6181	Interference Control Requirements, Aircraft Equipment
MIL-H-15362	Harness, Test Bench, Avionic Equipment, General Specification for
MIL-E-17555	Electronic and Electrical Equipment and Associated Repair Parts, Preparation for Delivery of
MIL-T-18303	Test Procedures, Preproduction and Inspection, for Aircraft Electronic Equipment, Format for
MIL-N-18307	Nomenclature and Nameplates for Airborne Electronic and Associated Equipment
MIL-T-21200	Test Equipment for Use with Electronic and Fire Control Systems, General Specification for

STANDARDS
Military

MIL-STD-781	Reliability Tests, Exponential Distribution

DRAWINGS
Bureau of Ships

63A114	Electronic Workbench Assembly

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2.1.1 Availability of Documents - When requesting specifications, standards, drawings, and publications, refer to both title and number. Copies of this specification and applicable specifications required by contractors in connection with specific procurement functions may be obtained upon application to the Commanding Officer, Naval Supply Depot, Code 105, 5801 Tabor Avenue, Philadelphia, Pennsylvania 19120.

3. REQUIREMENTS

3.1 Parts and Materials - In the selection of parts and materials, fulfillment of major design objectives shall be the prime consideration. In so doing, the following shall govern:

(a) Parts and materials requirements conforming to MIL-T-21200.

(b) Nonrepairable subassemblies, as specified in MIL-T-21200, shall be used when practicable. The general size of the subassembly, and the amount of circuitry to be included therein shall be approved by the procuring activity. Nonrepairable subassemblies must be reliable. A mean-time-to failure of more than 5000 hours should be the goal of each.

(c) When previously produced models of this equipment did not use nonrepairable subassemblies, the design shall not be changed to employ nonrepairable assemblies without the approval of the procuring activity.

3.2 Design and Construction

3.2.1 General - The equipment shall fullfill the applicable design, construction, and workmanship requirements as specified in MIL-H-15362, unless otherwise specified herein.

3.2.2 Total Weight - The total weight of the equipment shall not exceed 475 pounds.

3.2.3 Reliability

3.2.3.1 Operational Stability - The equipment shall operate with satisfactory performance, continuously or intermittently, for a period of 24 hours without necessity for readjustment of any operator controls which are inaccessible to the operator during use, except where adjustments associated with parts failures are required following replacement of those parts. Calibration every 30 days is required.

3.2.3.2 Operating Life - The equipment shall have a minimum total operating life of 2000 hours with reasonable servicing and replacement of parts.

3.2.3.3 Mean Time Between Failures (MTBF) - When maintained in accordance with applicable preventive maintenance procedures, the equipment shall demonstrate a specified MTBF of at least 200 hours. Accept-reject criteria shall be as specified in MIL-STD-781.

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3.2.4 Standard Conditions - The following conditions shall be used to establish normal performance requirements and for making laboratory bench tests:

Temperature	Room Ambient (30° $\pm 10^{o}$C)
Altitude	Normal Ground
Vibration	None
Humidity	Room Ambient up to 90% relative humidity
Input Power	115/200 ±6 VAC, 400 ±20 CPS, 3 PH and 27 ±2 VDC

3.2.5 Service Conditions - The equipment shall operate satisfactorily under any of the environmental service conditions, or reasonable combination of these conditions, as specified in MIL-H-15362. Sand, dust, fungus, explosion, and crash safety tests shall not be applicable to this equipment.

3.2.6 Installation - The equipment shall be used primarily for installation and operation on benches as specified in drawing 63A114 but shall also be designed to pass through an opening 26 inches by 54 inches with rounded corners of 8 inches radius in accordance with MIL-H-15362.

3.2.7 Workshop Primary Power - The equipment shall utilize the following power:

Voltage	Frequency	Phase
27 ±1	DC	
115/200 ±6	400 ±20 cps	3

3.2.8 Nomenclature and Nameplates - Nomenclature assignment and nameplate approval for equipment identification shall be in accordance with MIL-N-18307.

3.2.9 Interchangeability - The equipment shall meet the interchangeability requirements as specified in MIL-T-21200. Unless otherwise specified, interchangeability shall be measured against a model supplied by the procuring activity.

3.2.10 Maintenance Provisions and Field Testing - Provisions for maintenance shall be as specified in MIL-T-21200. Specific test points and test facilities shall be provided to the greatest extent practicable for ease of field testing and maintenance.

3.2.11 Undervoltage Protection - The equipment shall not be damaged by operation at 98 volts and shall automatically resume normal operation when the voltage returns within limits.

3.3 Detail Requirements

3.3.1 Stand, Test MT-2579/APM-145A - The Test Stand shall meet the following requirements:

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3.3.1.1 Function - The Test Stand shall position the Radar Set Group OY-12/APQ-124, or OY-12A/APQ-124 to facilitate the adjustment, alignment, testing repair, and replacement of components. The stand shall be capable of cantilevering suspension of the radar, allowing free access to the chassis for ease of maintenance. The rack shall allow rotation of the radar approximately 180 degrees about its major axis, with provision to lock the rotation in intermediate positions. The rack can also be rotated 90 degrees toward the front of the bench to facilitate maintenance of the antenna and both sides of the radar simultaneously. Provision shall also be made for locking the rotation in the 0 degree and the 90 degree positions.

3.3.1.2 Form Factor - The over-all dimensions of this unit shall not exceed 31 inches by 32 inches by 28 inches.

3.3.2 Base Electrical Equipment (MT-4046A/APM-293) - The base shall meet the following requirements:

3.3.2.1 Function - This unit shall provide facilities for mounting the Indicator, Azimuth-Range IP-869/APQ-124, or IP-869A/APQ-124; Power Supply PP-4835/APQ-124; Control, Radar Set C-7326/APQ-124, or C-7326A/APQ-124; Indicator, Missile Release IP-1485/APQ; and Hand Control Simulator SM-214A/APQ-83. The Indicator, Azimuth-Range shall be capable of being positioned approximately ±15 degrees in azimuth, and a method shall be incorporated to lock the assembly in any intermediate position.

3.3.2.2. Form Factor - The overall dimensions of this unit in locked position shall not exceed 16 inches by 10 inches by 17 inches. In an unlocked position the maximum length dimension shall be 20 inches.

3.3.3. Base Electrical Equipment MT-4048/APM-293 - The base shall meet the following requirements:

3.3.3.1 Function - This unit shall provide facilities for mounting the Synchronizer, Electrical SN-412/APQ-124, or SN-412A/APQ-124 during radar tests and shall provide roolout capabilities for maintenance and alignment.

3.3.3.2 Form Factor - The overall dimensions of this unit in locked position shall not exceed 29 inches by 21 inches by 4 inches. In an extended, unlocked position the maximum length dimension shall be 60 inches.

3.3.4 Base, Electrical Equipment MT-4047/APM-293 - The base shall meet the following requirements:

3.3.4.1 Function - This unit shall provide facilities for mounting the Processor, Coherent Data MX-7762/APQ-124 during the radar test.

3.3.4.2 Form Factor - The overall dimensions of this unit in locked position shall not exceed 19 inches by 23 inches by 3 inches. In an unlocked position the maximum length shall be 23 inches.

3.3.5 Base Electrical Equipment MT-4049/APM-293 - The base shall meet the following requirements:

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3.3.5.1 Function - This unit shall provide facilities for mounting the Gyroscope Displacement CN-529/APQ-83 and Amplifier-Control Group OG-39/APQ-124, or OG-39A/APQ-124 using MT-2308/APM-145, during radar tests and shall provide roll-out capabilities for maintenance and alignment.

3.3.5.2 Form Factor - The overall dimensions of this unit in a locked position shall not exceed 29 inches by 21 inches by 4 inches. In an extended, unlocked position the maximum length dimension shall be 60 inches.

3.3.6 Panel, Electrical Equipment MT-4045/APM-293 - The panel shall meet the following requirements:

3.3.6.1 Function - This unit shall provide facilities for mounting the Synchronizer, Electrical SN-411/APQ-124 and the Attitude Simulator SM-213/APQ-83 during radar test.

3.3.6.2 Form Factor - The overall dimensions of this unit shall not exceed 20 inches by 2 inches by 7 inches.

3.3.7 Base, Hydraulic Pump MT-4050/APM-293 - The base shall meet the following requirements:

3.3.7.1 Function - This unit shall provide facilities for mounting the Hydraulic Hand Pump of the Lift, Transfer, Radar Set MX-8196/APM-293 during positioning and mounting of the Radar Set Group OY-12/APQ-124, or OY-12A/APQ-124 in the Test Stand.

3.3.7.2 Form Factor - The overall dimensions of this unit shall not exceed 9 inches by 4 inches by 5 inches.

3.3.8 Mounting MT-2307/APM-145 - The mounting shall meet the following requirements:

3.3.8.1 Function - This unit shall provide facilities for mounting the Missile Release Computer CP-473A/APQ-83, or CP-473B/APQ-83 during radar test.

3.3.8.2 Form Factor - The over all dimensions of the unit in locked position shall not exceed 13-5/8 inches by 4 inches. In an unlocked position, the length shall not exceed 17 inches.

3.3.9 Mounting MT-2308/APM-145 - The mounting shall meet the following requirements:

3.3.9.1 Function - This unit shall provide facilities for mounting the Amplifier-Control Group OG-39/APQ-124, or OG-39A/APQ-124.

3.3.9.2 Form Factor - The overall dimensions of the unit shall not exceed 11-3/4 inches by 1-1/2 inches by 17-1/2 inches.

3.3.10 Mounting MT-3249/APM-145B - The mounting shall meet the following requirements:

3.3.10.1 Function - This unit shall provide facilities for mounting the Deviated Pursuit Computer CP-742A/APQ during radar test.

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3.3.10.2 Form Factor - The overall dimensions of this unit shall not exceed 5-3/4 inches by 11-1/2 inches by 3-3/4 inches.

3.3.11 Maintenance Fixture Electronic MT-3973/ASA-63 - The mounting shall meet the following requirements:

3.3.11.1 Function - This unit shall provide facilities for mounting the Programmer, Missile Acquisition AN/ASA-63, or AN/ASA-63A during radar test.

3.3.11.2 Form Factor - The overall length of this unit in a locked position shall not exceed 8 inches by 2 inches. In an unlocked position the length shall not exceed 14 inches.

3.3.12 Interconnecting Box J-2826A/APM-293 - The Interconnecting Box shall meet the following requirements:

3.3.12.1 Function - This unit shall provide the necessary circuitry, switches, circuit breakers, and interconnections for testing the Radar Set AN/APQ-124, or AN/APQ-124A. The associated cabling shall be point to point, and the interconnection of the Radar Set shall be accomplished within the Interconnecting Box. The exterior marking of the Interconnecting Box shall be accomplished by the use of silk screening. The cable marking shall be in accordance with MIL-T-21200.

3.3.12.2 Form Factor - The Interconnecting Box shall be surface-mounted and shall not exceed 14 inches by 14 inches by 9 inches.

3.3.13 Hose Assembly, Hydraulic MX-3012/APM-145 - The hose assembly shall provide a means of transmitting hydraulic power from the hydraulic supply to the bench top. Two of these hoses shall be required for each installation. This hose shall not exceed 50 inches in length.

3.3.14 Hose Assembly, Hydraulic MX-3571/APM-145A - The hose assembly shall provide the hydraulic power return line from Radar Set Group OY-12/APQ-124, or OY-12A/APQ-124 to the Stand, Test, MT-2579/APM-145A. This hose shall not exceed 36 inches in length.

3.3.15 Hose Assembly, Hydraulic MX-3572/APM-145A - The hose assembly shall provide the hydraulic power pressure line from the Stand, Test, MT-2579/APM-145A to the Radar Set Group OY-12/APQ-124, or OY-12A/APQ-124. This hose shall not exceed 30 inches in length.

3.3.16 Lift, Transfer, Radar Set MX-8196/APM-293 - The Lift, Transfer Radar Set MX-8196/APM-293 shall meet the following requirements:

3.3.16.1 Function - This unit shall lift and position the Radar Set Group OY-12/APQ-124, or OY-12A/APQ-124 during mounting in the Test Stand. The lift shall be a self-contained hydraulic and mechanical device. It shall contain a mechanical drive mechanism for axial traverse, a hydraulic cylinder and hand pump for vertical traverse, and shall provide a means for rotating the boom 90 degrees. The boom rotation shall have provisions to be locked in place at a stowage position, a position over the test stand, and a position approximately 90 degrees from the stowage position. The unit shall provide a Lift Bar, Radar Set to attach to the Radar Set Group OY-12/APQ-124, or OY-12A/APQ-124 with cover in place, for lifting and positioning the Radar

3.3.16.2 Form Factor - The unit shall mount on the test bench and the mounting dimension shall not exceed 43 inches by 31 inches by 28 inches. The boom extended shall not exceed 75 inches and retracted shall not exceed 52 inches.

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3.3.17 Adapter Plate, Radar Antenna Control MX-4844/APM-197 - The unit shall meet the following requirements:

3.3.17.1 Function - This unit shall provide the necessary electrical and mechanical facilities for operation of the Aircraft Radar Control during radar test.

3.3.17.2 Form Factor - The overall dimensions of this unit shall not exceed 8-1/2 inches by 9-1/2 inches by 6-1/2 inches.

3.3.18 Mount, Gyroscope MT-2093/APQ-83 - The unit shall meet the following requirements:

3.3.18.1 Function - This unit shall provide the necessary electrical and mechanical facilities for mounting the Gyro Assembly CN-529/APQ-83 during radar test.

3.3.18.2 Form Factor - The overall dimensions of this unit shall not exceed 7-1/2 inches by 8 inches by 2-1/2 inches.

4. QUALITY ASSURANCE PROVISIONS

4.1 Responsibility for Inspection - Unless otherwise specified in the contract or purchase order, the supplier is responsible for the performance of all inspection requirements as specified herein. Except as otherwise specified, the supplier may utilize his own facilities or any commercial laboratory acceptable to the Government. The Government reserves the right to perform any of the inspections set forth in the specification, where such inspections are deemed necessary to assure supplies and services conform to prescribed requirements.

4.1.1 Classification of Tests - Items covered by this specification shall be subjected to the following tests to determine compliance with all applicable requirements:

(a) Preproduction Tests
(b) Initial Production Tests
(c) Acceptance Tests
(d) Life Tests

4.1.1.1 Preproduction Tests - Preproduction tests shall be made on equipment representative of the production units to be supplied under the contract. Preproduction tests shall be accomplished under the responsibility of the contractor and shall be conducted in accordance with the approved test procedure of 4.1.2. The Government Inspector and the procuring activity shall be advised when tests are to be conducted, so that a representative may be designated to witness or supervise the tests, when so desired. Contractors not having adequate facilities to conduct all required tests shall obtain the services of a commercial testing laboratory acceptable to the Government.

4.1.1.1.1 Preproduction Test Data - The contractor shall submit all data collected in conducting these tests to the procuring agency for review and approval. On first production contracts this data shall include a list of all electrical and electronic parts, giving their specified voltage and their ambient and surface temperatures. The ambient and surface temperature shall be obtained under the extreme high temperature operating condition.

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4.1.1.1.2 Scope of Tests - Preproduction tests shall include all tests deemed necessary by the procuring activity to determine that the equipment meets all the requirements of this specification and the contract. Preproduction tests shall include environmental tests in accordance with MIL-T-21200 and interference tests in accordance with MIL-I-6181. Unless specified by contract action, sand and dust, fungus, or explosion tests will not be required.

4.1.1.1.3 Preproduction Approval - Approval of the preproduction sample equipment shall be by the procuring activity upon satisfactory completion of all tests. No production equipments shall be delivered prior to the approval of the preproduction sample. Prefabrication of production equipment prior to the approval of the preproduction sample is at the contractor's own risk. The approved preproduction sample shall be retained by the contractor for his use in the fabrication and testing of equipment to be submitted for acceptance. The preproduction sample shall not be considered as one of the equipments under the contract.

4.1.1.1.4 Production Equipments - Equipments supplied under the contract shall in all respects, including design, construction, workmanship, performance, and quality, be equivalent to the approved preproduction sample. Each equipment shall be capable of successfully passing the same tests imposed on the preproduction sample. Evidence of non-compliance with the above shall constitute cause for rejection of production equipment and for equipments already accepted by the Government. It shall be the obligation of the contractor to make necessary corrections as approved by the procuring activity.

4.1.1.2 Initial Production Tests - One of the first ten production equipments shall be selected and sent, at the contractor's expense, to a designated Government laboratory for tests. These equipments shall be selected by the procuring activity, after the quipment has successfully passed all individual tests. The preproduction sample shall not be selected for this test.

4.1.1.2.1 Scope of Tests - The equipments may be subjected to any and all tests the procuring activity deems necessary to assure that the production equipment is equivalent in design, construction, workmanship, performance, and quality to the previously approved preproduction sample and that it meets all applicalbe requirements.

4.1.1.2.2 Accessory Material - In addition to the complete equipment submitted for Initial Production Tests, the contractor shall also submit such accessory material and data necessary to test the equipment.

4.1.1.2.3 Initial Production Sample Approval - Approval of the Intitial Production Sample shall be by the procuring activity upon satisfactory completion of all tests. Any design, material, or performance defect made evident during this test shall be corrected by the contractor, to the satisfaction of the procuring activity. Failure of the Initial Production Sample to pass any of the tests shall be cause for deliveries of equipment under the contract to cease until proper corrective action is approved and accomplished. Corrective action shall also be accomplished on equipment previously accepted, when requested by the procuring activity.

4.1.1.2.4 Reconditioning of Initial Production Test Sample - On completion of the initial production test, the equipment shall be reworked by the contractor by replacing all "wear" or damaged items. After reworking, the contractor shall resubmit the equipment for acceptance.

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4.1.1.3 Acceptance Tests - The contractor shall furnish all samples and shall be responsible for accomplishing the acceptance tests. All inspection and testing shall be under the supervision of the government inspector. Contractors not having testing facilities satisfactory to the procuring activity shall engage the service of a commercial testing laboratory acceptable to the procuring activity. The contractor shall furnish test reports showing quantitative results for all acceptance tests. Such reports shall be signed by an authorized representative of the contractor or laboratory, as applicable. Acceptance or approval of material during the course of manufacture shall not be construed as a guarantee of the acceptance of the finished product. Acceptance tests shall consist of the following:

(a) Individual Tests

(b) Sampling Tests

(c) Reliability Assurance Tests

(d) Special Tests

4.1.1.3.1 Individual Tests - Each equipment submitted for acceptance shall be subjected to the individual tests. These tests shall be adequate to determine compliance with the requirements of material, workmanship, operational adequacy, and reliability. As a minimum, each equipment accepted shall have passed the following tests:

(a) Examination of Product

(b) Operational Test

(c) Manufacturing Run-in Test

4.1.1.3.1.1 Examination of Product - Each equipment shall be examined carefully to determine that the material and workmanship requirements have been met.

4.1.1.3.1.2 Operational Test - Each equipment shall be operated long enough to permit the equipment temperature to stabilize and to check sufficient characteristics and record adequate data to assure satisfactory operation.

4.1.1.3.1.3 Manufacturing Run-In Test - Each equipment shall be operated under the conditions specified herein for a period of 6 hours without failure. A failure shall be defined as anything which causes malfunctioning of the equipment. Only those adjustments which can be made by using such controls and adjustments that are accessible to the operator during the normal use of the equipment will be permitted.

Temperature	Ambient Room
Humidity	Ambient Room
Vibration	Any selection frequency within the range of 20 to 30 cps (excluding resonant points) and a minimum amplitude of ±3 g's.

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The equipment shall be vibrated (without vibration isolators) for a period of 10 minutes prior to the beginning of the 6-hour period of operation. Where feasible, the equipment shall be operated during this vibration period, for the purpose of detecting flaws and imperfect workmanship. Operation within the specified limits of satisfactory performance is not necessarily required during the vibration period. The direction of vibration should be vertical to the normal mounting plane for 5 minutes and lateral to the plane for 5 minutes. Where it is not feasible to vibrate the equipment in 2 directions, the vertical direction shall be used. During the 6-hour period of operation following the 10-minute vibration period, the equipment shall be mechanically cycled periodically through its various phases of operation. Should a failure occur, it should be repaired and the test started over, except that the 10-minute vibration period need not be repeated when it is certain the failure was not a result of the vibration. Should repetitive failures occur, corrective action shall be taken to eliminate this defect from future equipments. A record shall be kept of all failures. The 6-hours period specified above may be composed of two 3-hour periods to conform with standard working hours.

4.1.1.3.2 Sampling Tests - Equipments selected for sampling tests shall first have passed the individual tests. Equipments shall be selected for sampling tests by the government inspector in accordance with the following:

Quantity of Equipment Offered for Acceptance	Quantity to be Selected For Sampling Test
First 10	1
Next 50	1
Next 75	1
Next 100	1
for each additional 200 or fraction thereof	1

4.1.1.3.2.1 Scope of Tests - As a minimum, each equipment selected for sampling tests shall be subjected to the following tests:

(a) Complete operational test at ambient room conditions, making all necessary measurements to assure that all applicable specification requirements have been met.

(b) Operational test at certain environmental conditions. The conditions may vary for each equipment tested and should be based on results of the individual and special preproduction tests.

(c) Manufacturing run-in test specified in 4.1.1.3.1.3, except that the test duration shall be 120 hours with no restriction on the number of failures. However, each failure shall be analyzed as to cause and remedial action necessary to reduce the possibility of its recurrence in future equipments.

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4.1.1.3.3 Reliability Assurance Tests - Reliability Assurance Tests are not required unless specified by the contract.

4.1.1.3.4 Special Tests - Special tests shall be conducted on a quantity of equipments for the purpose of checking the effect of any design or material change on the performance of the equipment and to assure adequate quality control. The equipment selected for special tests may be selected from equipments previously subjected to the sampling or reliability assurance tests.

4.1.1.3.4.1 Special Test Schedule - Selection of equipment for special tests shall be made as follows:

(a) On an early equipment after an engineering or material change.

(b) Whenever failure reports or other information indicate additional tests are required. (This will be determined by the procuring activity.)

4.1.1.3.4.2 Scope of Tests - Special tests shall consist of such tests as approved by the procuring activity. Test procedures previously approved for the preproduction tests shall be used where applicable. When not applicable, the contractor shall prepare a test procedure and submit it to the procuring activity for approval prior to conducting the tests.

4.1.1.3.5 Equipment Failure - Should a failure occur during the sampling, reliability assurance, or special tests, the following action shall be taken:

(a) Determine the cause of failure.

(b) Determine if the failure is an isolated case or design defect.

(c) Submit to the procuring activity for approval, proposed corrective action intended to reduce the possibility of the same failure(s) occurring in future tests.

(d) Where practical, include a step within tests in the individual test to check all equipments for this requirement until reasonable assurance is obtained that the defect has been satisfactorily corrected.

4.1.1.4 Life Test - The contractor shall furnish all samples and shall be responsible for accomplishing the life test. The test shall be of 300-hours duration and shall be conducted on equipments that have passed the individual tests. The life test sample shall be performed under the conditions specified in 4.1.1.4.1. The life test sample shall be selected by the government inspector in accordance with the following. (Equipments which have successfully passed the Initial Production Test. Sampling Tests, Reliability Tests, or Special Tests may be selected for life tests.) When reliability tests are conducted the life test may be omitted if, during the reliability tests, a quantity of equipments equal to, or more than, that listed below receive at least 300 hours each of test time.

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7.2.8 Calibrator

Quantity of Equipments Offered for Acceptance	Quantity to be Selected for Life Test
First 25	1
Next 175	1
Next 300	1
for each additional 500 or fraction thereof	1

4.1.1.4.1 Test Conditions - The life test shall be conducted under the following simulated service conditions:

Temperature	Room Ambient (30° $\pm 10^{o}$C)
Altitude	Normal ground (0 - 5000 ft.)
Humidity	Room ambient
Input Voltage	115 ±10% (at lowest applicable frequency) 27.0 ±2.0 volts DC
Vibration	None

4.1.1.4.2 Test Periods - The test may be run continuously or intermittently. Any period of operation shall be of sufficient duration to permit the equipment temperature to stabilize. Periodically, the equipment shall be turned on and off several times and put through its various phases of operation.

4.1.1.4.3 Performance Check - At approximately 8-hour intervals during the test, a limited performance check shall be made. The performance check proposed by the contractor shall be subject to approval by the procuring activity.

4.1.1.4.4 Test Data - The contractor shall keep a daily record of the performance of the equipment, making particular note of any deficiences or failures. In the event of part failures, the defective part shall be replaced and the operation resumed for the balance of the test period. A record shall be kept of all failures throughout the test, including all tube failures. This record shall indicate the following:

(a) Part type number

(b) The circuit reference symbol number

(c) The part function

(d) Name of the manufacturer

(e) Nature of the failure

(f) The number of hours which the part operated prior to failure.

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4.1.1.4.4.1 Failure Report - In the event of a failure, the government inspector shall be notified immediately upon completion of the test. A report shall be submitted to the procuring activity. In this report, the contractor shall propose suitable and adequate design or material corrections for all failures which occurred. The procuring activity will review such proposals and determine whether they are acceptable.

4.1.1.4.5 Reconditioning of Life Test Samples - Equipment which has been subjected to the life test shall be reconditioned as follows:

(a) On completion of the life test, the equipment shall be reworked by the contractor by replacing all "wear" items. The "wear" items shall be determined by agreement between the contractor and the procuring activity.

(b) After reworking, the contractor shall resubmit the equipment for acceptance.

4.1.2 Test Procedures - The procedures used for conducting preproduction tests, acceptance tests, and life test shall be prepared by the contractor and submitted to the procuring activity for review and approval. The right is reserved by the procuring activity for the government inspector to modify the tests or require any additional tests deemed necessary to determine compliance with the requirements of this specification or the contract. MIL-T-18303 shall be used as a guide for preparation of test procedures. When approved test procedures are available from previous contracts, such procedures will be provided and may be used, when their use is approved by the procuring activity. However, the right is reserved by the procuring activity to require modification of such procedures, including additional tests, when deemed necessary.

4.1.3 Reconditioning of Tested Equipment - The equipment which has been subjected to acceptance and life tests shall be reconditioned by the contractor by replacing all "wear" or damaged items. After rework is completed the contractor shall resubmit the equipment for acceptance.

4.1.4 Presubmission Testing - No item, part, or complete equipment shall be submitted by the contractor until it has been previously tested and inspected by the contractor and found to comply, to the best of his knowledge and belief, with all applicable requirements.

4.1.5 Rejection and Retest - The equipment which has been rejected may be reworked, or have parts replaced to correct the defects, and resubmitted for acceptance. Before resubmitting, full particulars concerning previous rejection and the action taken to correct the defects found in the original shall be furnished the government inspector.

5. PREPARATION FOR DELIVERY

5.1 General - All major units and parts of the equipment shall be preserved, packaged, packed, and marked for the level of shipment specified in the contract or order in accordance with MIL-P-17555.

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6. NOTES

6.1 Use - The equipment shall include items on position, lift, support and interconnect Radar Set Group OY-12/APQ-124 or OY-12A/APQ-124; Indicator, Azimuth-Range IP-869/APQ-124 or IP-869A/APQ-124: Control, Radar Set C-7326/APQ-124 or C-7326A/APQ-124; Amplifier Control Group OG-39/APQ-124 or OG-39A/APQ-124; Synchronizer, Electrical SN-411/APQ-124; Gyro, Displacement CN-529/APQ-83; Computer, Missile Release CP-473A/APQ-83 or CP-473B/APQ-83; Synchronizer, Electrical SN-412/APQ-124 or SN-412A/APQ-124; Power Supply PP-4835/APQ-124; Indicator, Missile Release IP-1485/APQ: Hand Control Simulator SM-214A/APQ-83 or Aircraft Radar Control; Aircraft Attitude Simulator SM-213/APQ-83; and associated support equipment during test, operation, and maintenance on the test bench.

NOTICE: When Government drawings, specifications, or other data are used for any purpose other than in connection with a definitely related Government procurement operation, the United States Government thereby incurs no responsibility nor any obligation whatsoever; and the fact that the Government may have formulated, furnished, or in any way supplied the said drawings, specifications, or other data is not to be regarded by implication or otherwise as in any manner licensing the holder or any other person or corporation, or conveying any rights or permission to manufacture, use, or sell any patented invention that may in any way be related thereto.

6.2 Performance Characteristics - Normal and limiting values of performance data shall be determined at input voltages of 27.5 ±.5 VDC and 115 ±1.0 VAC, 3-phase, 400 cps. These data are to be used in testing the equipment at installation points, for compliance with minimum acceptance standard of performance.

6.3 Performance Objectives - Mimimum size and weight, simplicity of operation, ease of maintenance, and an improvement in the performance and reliability of the specific functions beyond the requirements of this specification are objectives which shall be considered in the production of this equipment. Where it appears that a substantial reduction in size and weight or improvement in simplicity of design, performance, ease of maintenance, or reliability will result from the use of materials, parts, and processes other than as specified in MIL-T-21200, it is desired their use be investigated. When investigation shows advantages can be realized, a request for approval shall be submitted to the procuring activity for consideration. Each request shall be accompanied by complete supporting information.

6.4 Non-Repairable Subassemblies - As a general rule non-repairable subassemblies should be encapsulated or hermetically sealed. The number of connections internal to the subassembly should be held to a minimum. Detail parts tolerances and ratings should be so selected that the life of the subassembly is greater than that of a similar repairable one. With a few exceptions (such as high voltage power supplies) the non-repairable subassembly should evidence a Mean-Time-to-Failure greater than 5000 hours, and for many applications this figure must be nearer 50,000 hours.

6.5 Precedence of Documents - When the requirements of the contract, this specification, or applicable subsidieary specifications are in conflict, the following precedence shall apply:

(a) Contract - The contract shall have precedence over any specification.

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(b) This Specification - This specification shall have precedence over all applicable subsidiary specifications. Any deviation from this specification, or from applicable subsidiary specifications, shall be specifically approved in writing by the procuring activity.

(c) Referenced Specifications - Any referenced specification shall have precedence over all applicable subsidiary specifications referenced therein. All referenced specifications shall apply to the extent specified.

6.6 The complete type number shall be used on nameplates, shipping records, and instruction books, as applicable.

6.7 Ordering Data - Purchasers should exercise any desired options offered herein, and procurement documents should specify the following:

(a) Title, number, and date of this specification.

(b) Selection of applicable levels of packaging and packing (see 5.1).

6.8 Notation of Revisions - A bar in the margin denotes a change, made in the adjacent paragraph, from the previous version of the document. This has been done as a convenience only, and the Government assumes no liability whatsoever for any inaccuracies in these notations. Bidders and contractors are cautioned to evaluate the requirements of this document, based on the entire content as written, irrespective of the marginal notations and relationship to the last previous issue.

6.9 Associated Equipment - The equipment shall operate with the following associated equipment:

(a)	Adapter Plate Radar Antenna Control	MX-4844/APM-197
(b)	Simulator Hand Control	SM-214/APQ-83
(c)	Simulator Aircraft Attitude	SM-213/APQ-83
(d)	Pumping Unit Hydraulic - Bench	MX-3196/APQ-83
(e)	Test Set Radar	AN/UPM-290A

Project No. 4940-N306

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SPECIFICATION ANALYSIS SHEET

Form Approved
Budget Bureau No. 119-R004

INSTRUCTIONS

This sheet is to be filled out by personnel either Government or contractor, involved in the use of the specification in procurement of products for ultimate use by the Department of Defense. This sheet is provided for obtaining information on the use of this specification which will insure that suitable products can be procured with a minimum amount of delay and at the least cost. Comments and the return of this form will be appreciated. Fold on lines on reverse side, staple in corner, and send to preparing activity *(as indicated on reverse hereof)*.

SPECIFICATION

MIL-T-81759A(AS) TEST HARNESS, RADAR, AN/APM-293A

ORGANIZATION *(Of submitter)*	CITY AND STATE

CONTRACT NO.	QUANTITY OF ITEMS PROCURED	DOLLAR AMOUNT
		$

MATERIAL PROCURED UNDER A

☐ DIRECT GOVERNMENT CONTRACT ☐ SUBCONTRACT

1. HAS ANY PART OF THE SPECIFICATION CREATED PROBLEMS OR REQUIRED INTERPRETATION IN PROCUREMENT USE?

A. GIVE PARAGRAPH NUMBER AND WORDING.

B. RECOMMENDATIONS FOR CORRECTING THE DEFICIENCIES.

2. COMMENTS ON ANY SPECIFICATION REQUIREMENT CONSIDERED TOO RIGID

3. IS THE SPECIFICATION RESTRICTIVE?

☐ YES ☐ NO IF "YES", IN WHAT WAY?

4. REMARKS *(Attach any pertinent data which may be of use in improving this specification. If there are additional papers, attach to form and place both in an envelope addressed to preparing activity)*

SUBMITTED BY *(Printed or typed name and activity)*	DATE

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Naval Air Systems Command
Washington, D.C. 20360

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